

Fourth Semester B. Tech. (Electronics and Communication Engineering) Examination

ANALOG CIRCUITS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Assume suitable data and illustrate answer with sketches wherever necessary.
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1.
 - (a) Develop the expression for input resistance of voltage series feedback amplifier and current series feedback amplifier. 5(CO1,2)
 - (b) An amplifier has mid-band voltage gain ($A_{v \text{ mid}}$) of 1000 with $f_L = 50 \text{ Hz}$ and $f_H = 50 \text{ kHz}$, if 5% feedback is applied then calculate gain f_L and f_H with feedback. 5(CO1,2)
 2.
 - (a) Develop the expression for the maximum efficiency of class-B push pull amplifier. 5(CO3)
 - (b) What is a cross-over distortion ? Explain how the harmonic distortion in push-pull amplifier is reduced. 5(CO3)
 3.
 - (a) What is Barkhausen criterion for the feedback oscillator ? 1(CO2,3)
 - (b) Explain the working of Wein bridge oscillator. Develop the formula for the frequency. 6(CO2,3)
 - (c) In a transistorized Hartley oscillator the two inductances are 2 mH and 20 μH while the frequency is to be changed from 950 kHz to 2050 kHz. Calculate the range over which the capacitor is to be varied. 3(CO2,3)

4. (a) For the cascaded differential amplifier shown in Fig. 1, following specifications are given $R_C = 8.6 \text{ k}\Omega$, $R_1 = 10 \text{ k}\Omega$, $V_T = 26 \text{ mV}$, $\beta_{dc} = 100$ typical, $\beta_{ac} = 100$, $V_{CC} = 10 \text{ V}$, $-V_{EE} = -10 \text{ V}$ and $V_{BE} = 0.7 \text{ V}$ evaluate :
- The collector current and collector to emitter voltage for each transistor.
 - The overall voltage gain.
 - The input resistance.
 - The output resistance.

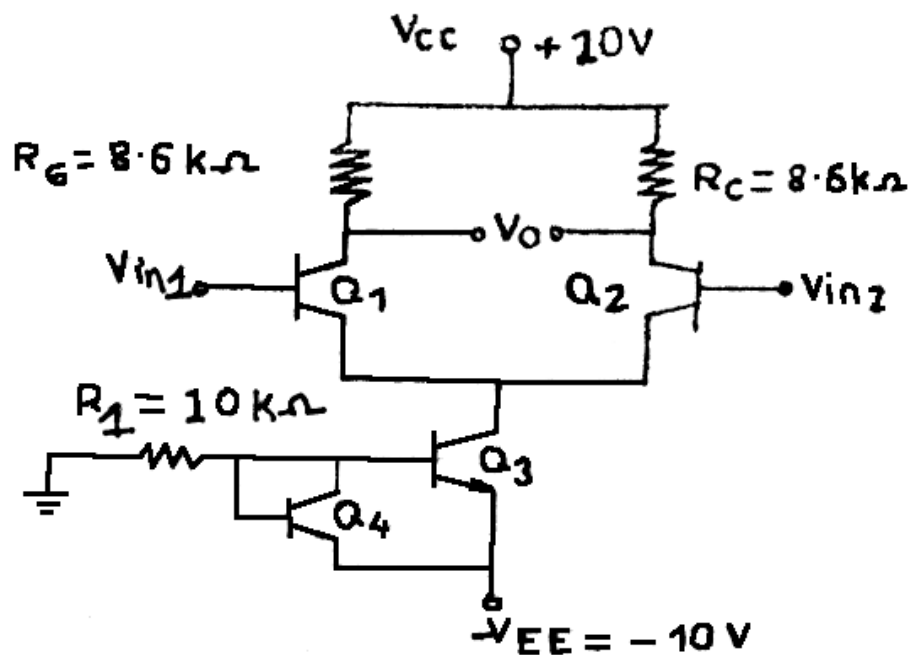


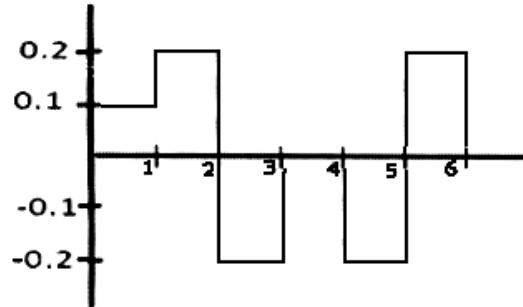
Fig. 1

5(CO3)

- (b) Design dual input balanced output differential amplifier using diode current bias to meet the following specifications :
- Supply voltages $V_s = \pm 12 \text{ V}$.
 - Emitter current I_E in each differential amplifier 1.5 mA .
 - Voltage gain = 60.

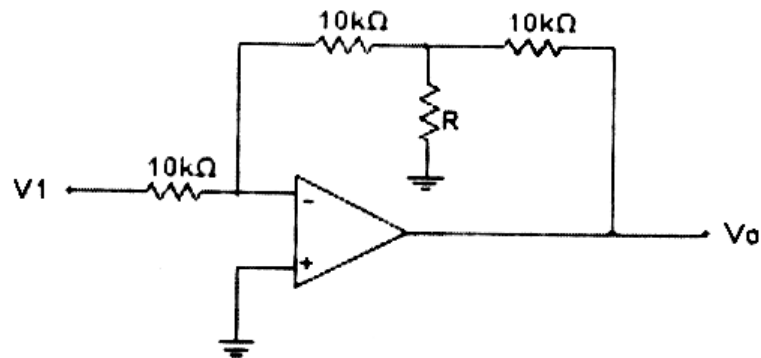
5(CO3)

5. (a) For the basic integrator circuit if the input is as shown in the Figure below and $R_1C_f = 100$ msec, determine the output and sketch it. Assume that the op-amp is initially nulled.



5(CO4)

- (b) For the circuit shown in Figure below, find the value of R required to obtain, $V_o = -50 V_1$.



5(CO4)

6. (a) Design 4th Order Butterworth Low Pass filter with $F_h = 1$ KHz. 5(CO5)
- (b) Design an astable multivibrator using IC 555 having output frequency of 10 KHz and duty cycle 25%. 5(CO5)

